

ADESH PARTAP SINGH

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LinkedIn 

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EDUCATION

Bachelors of Applied Science, Mechatronics Engineering University of Waterloo

September 2021 - May 2026

- Computational Intelligence - ML
- Power Electronics and Motor Drives
- Mechatronic System Integration - PLC's
- Digital Control Applications
- Microprocessor Systems
- Algorithm Design & Analysis
- Operating Systems
- Embedded Computer Systems

WORK EXPERIENCE

Tesla - Controls Engineering Intern, San Diego

May 2025 - August 2025

- Developed a custom HMI and engineered a TCP/IP-based data transmission system in Python, allowing remote label printing across the Tesla network, streamlining manufacturing operations.
- Programmed PLCs in **Siemens TIA** Portal V18 to process real-time sensor data and visualize heat exchange rate and torque percentage using **Grafana** dashboards.
- Redesigned the recipe editor to incorporate advanced features including torque calculation, sliding window average/max/min analysis on live data, and an integrated alarm system for improved process monitoring.

Motorola Solutions - Software Engineering Intern, Concord

January 2025 - April 2025

- Implemented new features using C++ to an internal dashboard by adding real-time radio parameters, configuration features, enabling technicians to improve communication reliability in field units
- Conducted network vulnerability scans using **Nessus** and Tenable.io to identify open ports, insecure gateways, and potential threats prior to deployment of communication devices used by law enforcement.

Neutron Controls - Embedded Systems Intern, Mississauga

May 2024 - August 2024

- Prototyped the company's first Battery Passport system, enabling secure logging of cell-level data using **SEMPER NOR Flash** and Infineon TLE9018 via **SPI** communication.
- Engineered a custom LED indicator board to visually display real-time charge/discharge cycles on the BMS.
- Supported the BMS architecture upgrade from 12 to 18 cells, enhancing overall energy capacity.

Accelerated Systems Inc (ASI) - PCB Hardware Designer, Waterloo

September 2023 - December 2023

- Designed multi-layered boards like HV Buck convertor, DC Motor Controller, and Gyro Stabilizer.
- Simulated analog circuits using **LTspice** and **MATLAB** Simulink; tested and tuned motor performance using ASI's proprietary BACDoor software, providing actionable optimization insights.

ADDITIONAL INFORMATION

Relevant Projects & Skills

- Stewarts Platform - PID Control, UART, I2C, Inverse Kinematics
- STM32 and other PCB's - Altium
- FPGA - VHDL, RTL programming
- JARVIS Chatbot - openai library, custom RAG capability, text to voice and vice versa capability
- Monte Carlo simulation - BSM Model to calculate price of call/put options
- Facial Recognition attendance system - OpenCV
- Stock Trading & Poker Bot - simulated real time outcomes using the best possible algorithms

Electrical Team Lead - Waterloo

- Spearheaded key projects such as the Levitating Pod, Motor Control Unit, & Exterior Lighting for 6th Gen Pod
- Bought in funding of **\$5000** by pitching our Team to UWaterloo Finance Association, and other local groups.

Digiresfolio.com

- Found a startup for University of Waterloo students enabling online portfolio creation; achieved 100+ user signups within the first month, helping students showcase their technical projects.

Tools: SCADA Ignition, Siemens TIA Portal, TwinCAT, Tasking, Altium, KiCAD, Matlab, Proteus, Figma, Jira

Languages: Python, C/C++, Verilog, JavaScript, HTML, CSS